

PROJECT NAME:

Healthcare Patient Monitoring System.



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INTRODUCTION

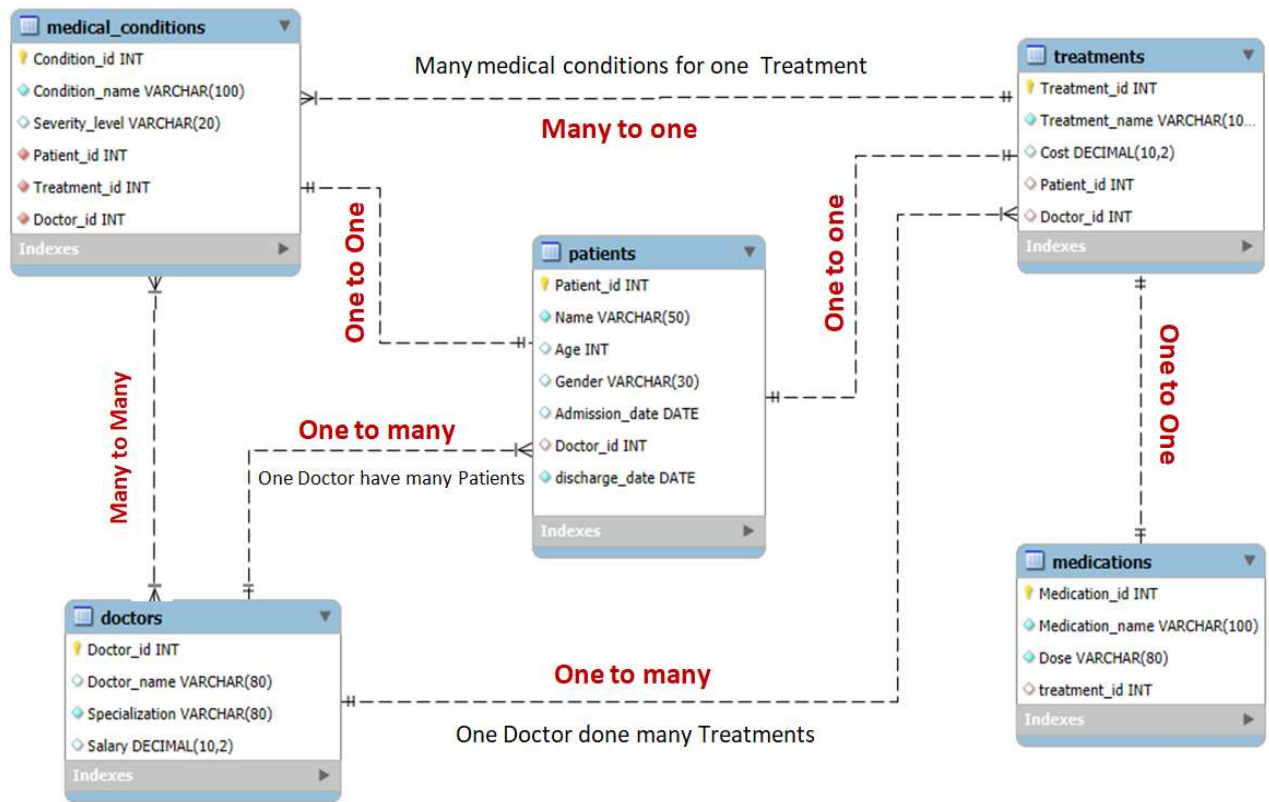
The **Healthcare Patient Monitoring System** is designed to efficiently track and manage vital patient information, including medical history, ongoing treatments, and doctor assignments. This system is an SQL-based project that simulates a real-world hospital environment, focusing on storing and retrieving data related to patients, doctors, treatments, medications, and medical conditions.

The primary goal of the system is to provide healthcare professionals with quick access to essential patient information, enabling better decision-making and improved patient care. By maintaining structured records of patient details, treatment plans, and doctor assignments, the system ensures that critical healthcare data is well-organized and readily available.

From a technical perspective, the project includes the creation of relational tables, views, and queries that streamline data management. The project leverages SQL to perform data manipulation and query operations such as filtering, sorting, aggregating, and combining data to gain insights. It also employs advanced SQL concepts like subqueries, joins, and views to enhance the functionality of the database.

Additionally, the project is relevant to my Statistics background, as it involves data analysis, pattern identification, and insights generation from medical and patient data. Statistical concepts are naturally integrated into the project's queries, enabling the analysis of healthcare trends, patient demographics, treatment costs, and doctor-patient relationships. This project not only strengthens my technical SQL skills but also demonstrates how statistical thinking can improve healthcare data analysis and decision-making.

ER DIAGRAM

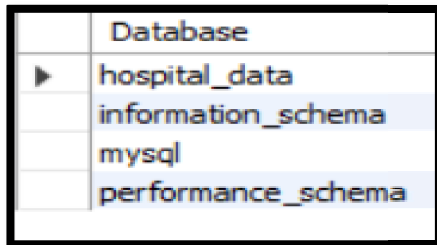


Databases :

Create Database Hospital_data

Use Hospital_data

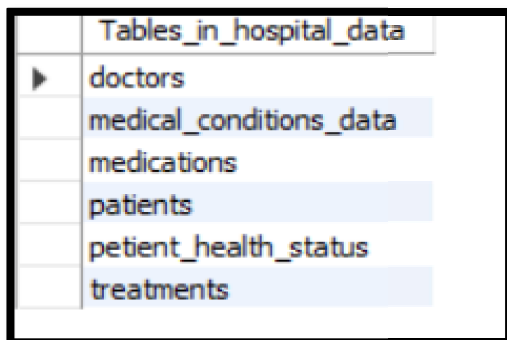
Show databases;

A screenshot of a MySQL command-line interface showing the output of the 'SHOW DATABASES;' command. The output is a table with two columns: a selection checkbox and the database name. The databases listed are hospital_data, information_schema, mysql, and performance_schema. The 'hospital_data' row is highlighted with a blue background.

	Database
☒	hospital_data
☐	information_schema
☐	mysql
☐	performance_schema

Tables in Hospital_data Database :

show tables;

A screenshot of a MySQL command-line interface showing the output of the 'SHOW TABLES;' command within the 'hospital_data' database. The output is a table with two columns: a selection checkbox and the table name. The tables listed are doctors, medical_conditions_data, medications, patients, patient_health_status, and treatments. The 'doctors' row is highlighted with a blue background.

	Tables_in_hospital_data
☒	doctors
☐	medical_conditions_data
☐	medications
☐	patients
☐	patient_health_status
☐	treatments

1.DATA DEFINITION LANGUAGE (DDL):

1.Creating Tables :

A)Patients

Create table Patients(Patient_id int primary key,Name varchar(50) not null,Age int check (age>0),Gender varchar(30),Admission_date date,Doctor_id int);

Desc patients;

	Field	Type	Null	Key	Default	Extra
►	Patient_id	int	NO	PRI	NULL	
	Name	varchar(50)	NO		NULL	
	Age	int	YES		NULL	
	Gender	varchar(30)	YES		NULL	
	Admission_date	date	YES		NULL	
	Doctor_id	int	YES	MUL	NULL	

B)Doctors

Create table Doctors(Doctor_id Int primary key,Doctor_Name varchar(50) not null,Specialization varchar(80) not null,Salary Decimal(10,2) check (salary>0));

Desc Doctors;

	Field	Type	Null	Key	Default	Extra
►	Doctor_id	int	NO	PRI	NULL	
	Doctor_Name	varchar(50)	NO		NULL	
	Specialization	varchar(80)	NO		NULL	
	Salary	decimal(10,2)	YES		NULL	

C)Treatments

Create table Treatments(Treatment_id Int Primary key not null,name varchar(100) not null,Cost decimal(10,2),Patient_id int,Doctor_id int);

Desc Treatments;

	Field	Type	Null	Key	Default	Extra
►	Treatment_id	int	NO	PRI	NULL	
	name	varchar(100)	NO		NULL	
	Cost	decimal(10,2)	YES		NULL	
	Patient_id	int	YES	MUL	NULL	
	Doctor_id	int	YES	MUL	NULL	

D) Medications

Create table Medications(Medication_id int Primary key not null, Medication_name varchar(100) not null, Dose varchar(80) not null, Administration_Method varchar(100) not null, treatment_id int, foreign key(treatment_id) references treatments(treatment_id));

Desc Medications;

	Field	Type	Null	Key	Default	Extra
►	Medication_id	int	NO	PRI	NULL	
	Medication_name	varchar(100)	NO		NULL	
	Dose	varchar(80)	NO		NULL	
	Administration_Method	varchar(100)	NO		NULL	
	treatment_id	int	YES	MUL	NULL	

E) Medical_Conditions_data

Create table medical_conditions_data(
condition_id int not null auto_increment, condition_name varchar(100) not null,
severity_level varchar(20), patient_id int not null,
treatment_id int not null, doctor_id int not null,
foreign key (patient_id) references patients(patient_id),
foreign key (treatment_id) references treatments(treatment_id),
foreign key (doctor_id) references doctors(doctor_id),
primary key (condition_id, doctor_id) -- composite primary key for many-to-many);
desc medical_conditions_data;

	Field	Type	Null	Key	Default	Extra
►	Condition_id	int	NO	PRI	NULL	auto_increment
	Condition_name	varchar(100)	NO		NULL	
	Severity_level	varchar(20)	YES		NULL	
	Patient_id	int	NO	MUL	NULL	
	Treatment_id	int	NO	MUL	NULL	
	Doctor_id	int	NO	PRI	NULL	

F) Patient_Health_status

```
create table Patient_Health_status(Status_id int primary key,Petient_id int not null,Heart_rate Int,BP  
int check(BP>0));
```

```
desc patient_Health__status;
```

	Field	Type	Null	Key	Default	Extra
►	Status_id	int	NO	PRI	NULL	
	Petient_id	int	NO		NULL	
	Heart_rate	int	YES		NULL	
	BP	int	YES		NULL	

2.Alter table :

- **Alter Table : Add column**

```
alter table patients add column discharge_date date not null;
```

	Field	Type	Null	Key	Default	Extra
►	Patient_id	int	NO	PRI	NULL	
	Name	varchar(50)	NO		NULL	
	Age	int	YES		NULL	
	Gender	varchar(30)	YES		NULL	
	Admission_date	date	YES		NULL	
	Doctor_id	int	YES	MUL	NULL	
	discharge_date	date	NO		NULL	

- **Alter Table : Modify Column**

```
alter table Doctors modify Doctor_name varchar(80) ;
```

	Field	Type	Null	Key	Default	Extra
►	Doctor_id	int	NO	PRI	NULL	
	Doctor_name	varchar(80)	YES		NULL	
	Specialization	varchar(80)	NO		NULL	
	Salary	decimal(10,2)	YES		NULL	

- **Alter Table : Rename column**

```
alter table Treatments rename column name to Treatment_name;
```

	Field	Type	Null	Key	Default	Extra
►	Treatment_id	int	NO	PRI	NULL	
	Treatment_name	varchar(100)	NO		NULL	
	Cost	decimal(10,2)	YES		NULL	
	Patient_id	int	YES	MUL	NULL	
	Doctor_id	int	YES	MUL	NULL	

- **Alter Table : Drop column**

`alter table Medications drop column Administration_Method;`

	Field	Type	Null	Key	Default	Extra
►	Medication_id	int	NO	PRI	NULL	
	Medication_name	varchar(100)	NO		NULL	
	Dose	varchar(80)	NO		NULL	
	treatment_id	int	YES	MUL	NULL	

- **Alter Table : Rename table**

`alter table Medical_Conditions_data rename Medical_conditions;`

	Field	Type	Null	Key	Default	Extra
►	Condition_id	int	NO	PRI	NULL	auto_increment
	Condition_name	varchar(100)	NO		NULL	
	Severity_level	varchar(20)	YES		NULL	
	Patient_id	int	NO	MUL	NULL	
	Treatment_id	int	NO	MUL	NULL	
	Doctor_id	int	NO	MUL	NULL	

3.Truncate table :

`truncate Patient_Health_status;`

	Status_id	Patient_id	Heart_rate	BP
*	NULL	NULL	NULL	NULL

4.Drop Table :

`Drop table Patient_Health_status;`

2.DATA MANIPULATION LANGUAGE (DML):

1. Insert into table :

`Insert into Doctors(Doctor_id,Doctor_Name,Specialization,Salary) values (6,'Dr.Anushree Patil','Cardiology',120000.00);`

`select*from doctors;`

	Doctor_id	Doctor_name	Specialization	Salary
▶	1	Dr.Anushree Patil	Cardiology	120000.00
	2	Dr.Sameer Shaikh	Neurology	185000.00
	3	Dr.Sushant Davis	Orthopedics	100000.00
	4	Dr.Aradhya Patel	Oral surgery dentist	100000.00
	5	Dr.Manas Devkar	General Practitioner	100000.00
	6	Dr.Anushree Patil	Cardiology	120000.00

2. Update into Table :

Q. Update Patient Doctor and Discharge date.

`update patients`

`set doctor_id =1, discharge_date = '2024-01-20'`

`where patient_id = 101;`

	Patient_id	Name	Age	Gender	Admission_date	Doctor_id	discharge_date
▶	101	Raj Kumar	45	Male	2024-01-08	1	2024-01-20
	102	Riya Patil	33	Female	2024-02-20	2	2024-02-28
	103	Priyanka More	40	Female	2024-02-25	3	2024-03-02
	104	Rajani Bapat	28	Female	2024-03-01	3	2024-03-16
	105	Pooja Mane	39	Female	2024-03-11	5	2024-03-20
	106	Kumar Sharma	55	Male	2024-03-19	1	2024-03-30
	107	Alisha Patil	49	Female	2024-03-31	4	2024-04-04
	108	Shyam Devan	58	Male	2024-04-04	5	2024-04-16
	109	Karishma Sutar	24	Female	2024-04-03	4	2024-04-19

3. Delete from table :

Q.Delete record having doctor_id 6.

`Delete from Doctors where doctor_id=6;`

	Doctor_id	Doctor_name	Specialization	Salary
▶	1	Dr.Anushree Patil	Cardiology	120000.00
	2	Dr.Sameer Shaikh	Neurology	185000.00
	3	Dr.Sushant Davis	Orthopedics	100000.00
	4	Dr.Aradhya Patel	Oral surgery dentist	100000.00
	5	Dr.Manas Devkar	General Practitioner	100000.00
●	NULL	NULL	NULL	NULL

3.DATA QUERY LANGUAGE (DQL) :

1.Select Query:

a) Select Query for entire data.

Select*from Patients;

	Patient_id	Name	Age	Gender	Admission_date	Doctor_id	discharge_date
▶	101	Raj Kumar	45	Male	2024-01-08	1	2024-01-20
	102	Riya Patil	33	Female	2024-02-20	2	2024-02-28
	103	Priyanka More	40	Female	2024-02-25	3	2024-03-02
	104	Rajani Bapat	28	Female	2024-03-01	3	2024-03-16
	105	Pooja Mane	39	Female	2024-03-11	5	2024-03-20
	106	Kumar Sharma	55	Male	2024-03-19	1	2024-03-30
	107	Alisha Patil	49	Female	2024-03-31	4	2024-04-04
	108	Shyam Devan	58	Male	2024-04-04	5	2024-04-16
	109	Karishma Sutar	24	Female	2024-04-03	4	2024-04-19

b)Select specific data i.e Medication name and dose from table.

Select medication_name,dose from Medications;

	medication_name	dose
▶	Aspirin	120 mg
	Ibuprofen	400mg
	Celecoxib	200 mg
	Hyaluronic acid	80 mg
	Acetaminophen	500 mg
	Clopidogrel	75 mg
	Acetaminophen	500 mg
	Atorvastatin	70 mg
	Amoxicillin	500 mg

c)Select query with changing Column name.

Select treatment_name as treatment_for_Patients from treatments;

	treatment_for_Patients
▶	Heart Surgery
	MRI Scan for Headache
	Knee replacement
	Joint Injection
	Routine physical exam
	Angioplasty
	Tooth Extraction
	Health Screening
	Dental implants

2.Order by

a)List of doctors in ascending order by salary.

Select*from doctors order by salary;

	Doctor_id	Doctor_name	Specialization	Salary
▶	3	Dr.Sushant Davis	Orthopedics	100000.00
	4	Dr.Aradhya Patel	Oral surgery dentist	100000.00
	5	Dr.Manas Devkar	General Practitioner	100000.00
	1	Dr.Anushree Patil	Cardiology	120000.00
	2	Dr.Sameer Shaikh	Neurology	185000.00
*	NULL	NULL	NULL	NULL

b)List of medical conditions in descending order by treatment_id.

Select*from medical_conditions order by treatment_id desc;

	Condition_id	Condition_name	Severity_level	Patient_id	Treatment_id	Doctor_id
▶	9	Tooth Loss	Severe	109	209	4
	8	Health Checkup	Mild	108	208	5
	7	Tooth Decay	Moderate	107	207	4
	6	Coronary Artery Disease	Severe	106	206	1
	5	General Checkup	Mild	105	205	5
	4	Joint Pain	Mild	104	204	3
	3	Osteoarthritis	Severe	103	203	1
	3	Osteoarthritis	Severe	103	203	3
	2	Chronic Headache	Moderate	102	202	2
	1	Heart Disease	Severe	101	201	1
	1	Heart Disease	Severe	101	201	5

3.Limit Query

Display top 5 most expensive treatments.

Select*from Treatments order by cost desc limit 5;

	Treatment_id	Treatment_name	Cost	Patient_id	Doctor_id
▶	201	Heart Surgery	90000.00	101	1
	206	Angioplasty	82000.00	106	1
	203	Knee replacement	65000.00	103	3
	202	MRI Scan for Headache	25000.00	102	2
	209	Dental emplants	25000.00	109	4
*	NULL	NULL	NULL	NULL	NULL

4.Distinct Query

Display Unique Gender from Patients.

Select distinct gender from patients;

	gender
▶	Male
	Female

5.Where Clause:

1)With Comparison Operator

Find patients who were discharged on before March 31, 2024.

```
select * from patients
```

```
where discharge_date < '2024-03-31';
```

	Patient_id	Name	Age	Gender	Admission_date	Doctor_id	discharge_date
▶	101	Raj Kumar	45	Male	2024-01-08	1	2024-01-20
	102	Riya Patil	33	Female	2024-02-20	2	2024-02-28
	103	Priyanka More	40	Female	2024-02-25	3	2024-03-02
	104	Rajani Bapat	28	Female	2024-03-01	3	2024-03-16
	105	Pooja Mane	39	Female	2024-03-11	5	2024-03-20
	106	Kumar Sharma	55	Male	2024-03-19	1	2024-03-30
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2)Logical Operator

➤ Using AND Operator

Find patients who are older than 35 and are female.

```
select * from patients
```

```
where age > 35 and gender = 'female';
```

	Patient_id	Name	Age	Gender	Admission_date	Doctor_id	discharge_date
▶	103	Priyanka More	40	Female	2024-02-25	3	2024-03-02
	105	Pooja Mane	39	Female	2024-03-11	5	2024-03-20
	107	Alisha Patil	49	Female	2024-03-31	4	2024-04-04
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL

➤ Using AND/OR Operator

Find doctors who specialize in either 'Cardiology' or 'Neurology'.

```
Select * from doctors
```

```
where salary >= 100000 and specialization = 'cardiology' or specialization = 'neurology';
```

	Doctor_id	Doctor_name	Specialization	Salary
▶	1	Dr.Anushree Patil	Cardiology	120000.00
	2	Dr.Sameer Shaikh	Neurology	185000.00
✱	NULL	NULL	NULL	NULL

➤ **Using NOT Operator**

Find patients id,treatment and cost who are not treated by doctor with ID 3 & 4.

```
select patient_id,treatment_name,cost from treatments  
where not doctor_id = 3 and 4;
```

	Patient_id	treatment_name	cost
▶	101	Heart Surgery	90000.00
	106	Angioplasty	82000.00
	102	MRI Scan for Headache	25000.00
	107	Tooth Extraction	12000.00
	109	Dental emplants	25000.00
	105	Routine physical exam	8000.00
	108	Health Screening	12000.00

➤ **Using Not null**

Display details of medical_conditions where severity is not null.

```
Select*from Medical_conditions WHERE Severity_level is not null;
```

	Condition_id	Condition_name	Severity_level	Patient_id	Treatment_id	Doctor_id
▶	1	Heart Disease	Severe	101	201	1
	1	Heart Disease	Severe	101	201	5
	2	Chronic Headache	Moderate	102	202	2
	3	Osteoarthritis	Severe	103	203	1
	3	Osteoarthritis	Severe	103	203	3
	4	Joint Pain	Mild	104	204	3
	5	General Checkup	Mild	105	205	5
	6	Coronary Artery Disease	Severe	106	206	1
	7	Tooth Decay	Moderate	107	207	4
	8	Health Checkup	Mild	108	208	5
	9	Tooth Loss	Severe	109	209	4

➤ **Using BETWEEN operator**

Find treatments with costs between 10,000 and 50,000.

```
select*from treatments where cost between 10000 and 50000;
```

	Treatment_id	Treatment_name	Cost	Patient_id	Doctor_id
▶	202	MRI Scan for Headache	25000.00	102	2
	204	Joint Injection	18000.00	104	3
	207	Tooth Extraction	12000.00	107	4
	208	Health Screening	12000.00	108	5
	209	Dental emplants	25000.00	109	4

➤ Using IN operator

Find medical conditions that have either 'Severe' or 'Moderate' severity and are related to treatments with IDs 201, 203, or 206.

```
select * from medical_conditions
```

```
where severity_level in('severe','Moderate') and treatment_id in(201,203,206);
```

	Condition_id	Condition_name	Severity_level	Patient_id	Treatment_id	Doctor_id
▶	1	Heart Disease	Severe	101	201	1
	1	Heart Disease	Severe	101	201	5
	3	Osteoarthritis	Severe	103	203	1
	3	Osteoarthritis	Severe	103	203	3
	6	Coronary Artery Disease	Severe	106	206	1
*	NULL	NULL	NULL	NULL	NULL	NULL

➤ Using ANY operator

Find treatments and cost having cost higher than any treatment costing less than 10,000.

```
Select treatment_name, cost
```

```
from treatments
```

```
where cost > any (select cost from treatments where cost < 10000);
```

	Treatment_name	Cost
▶	Heart Surgery	90000.00
	MRI Scan for Headache	25000.00
	Knee replacement	65000.00
	Joint Injection	18000.00
	Angioplasty	82000.00
	Dental emplants	25000.00

➤ Using ALL operator

Find doctors whose salary is greater than the salary of all doctors with a specialization in 'General Practitioner'.

```
select doctor_name, specialization, salary
```

```
from doctors
```

```
where salary > all (select salary from doctors where specialization = 'general practitioner');
```

	Doctor_Name	Specialization	Salary
▶	Dr. Anushree Patil	Cardiology	120000.00
	Dr. Sameer Shaikh	Neurology	185000.00

6. Aggregate Functions:

➤ Count Function:

Find the total number of patients admitted.

Select count(*) as total_number_petients from patients;

	TOTAL_NUMBER_PETIENTS
▶	9

➤ Average Function with round function:

Find the average cost of all treatments upto two decimal places.

Select round(avg(cost) , 2) as average_treatment_cost
from treatments;

	Average_Treatment_Cost
▶	37444.44

➤ Sum Function :

Display total Salary Paid to All Doctors.

select sum(salary) as total_doctor_salary
from doctors;

	Total_Doctor_Salary
▶	605000.00

➤ Max, Min Function:

Find the highest and lowest salary of doctors.

Select max(salary) as highest_salary, min(salary) as lowest_salary
from doctors;

	Highest_Salary	Lowest_Salary
▶	185000.00	100000.00

7.Group by clause :

To display Total_treatment_fee for each doctor.

```
Select doctor_id,sum(cost) as total_treatment_fee  
from treatments  
group by doctor_id;
```

	doctor_id	Total_treatment_fee
▶	1	172000.00
	2	25000.00
	3	83000.00
	4	37000.00
	5	20000.00

Display gender wise Total Number_of_Patients.

```
select gender,count(patient_id) as number_of_patients  
from patients  
group by gender;
```

	Gender	Number_of_Patients
▶	Male	3
	Female	6

How many medical conditions are there for each severity level, and which patient has the earliest ID within each severity group.

```
select severity_level,count(condition_id) as number_of_conditions,min(patient_id) as  
earliest_patient  
from medical_conditions  
group by severity_level;
```

	Severity_level	Number_of_Conditions	Earliest_Patient
▶	Severe	6	101
	Moderate	2	102
	Mild	3	104

8.Having Clause:

Find Doctors with Total Treatment Costs Exceeding 50,000.

```
select doctor_id, sum(cost) as total_treatment_cost
from treatments
group by doctor_id
having total_treatment_cost > 50000;
```

	Doctor_id	Total_Treatment_Cost
▶	1	172000.00
	3	83000.00

9.Like Operator :

find treatments with names that have exactly 4 characters followed by the word 'scan'.

```
select * from treatments
where treatment_name like '____scan%';
```

	Treatment_id	Treatment_name	Cost	Patient_id	Doctor_id
▶	202	MRI Scan for Headache	25000.00	102	2
⋮	NULL	NULL	NULL	NULL	NULL

Find medical conditions that include 'disease' in their names.

```
select * from medical_conditions
where condition_name like '%disease%';
```

	Condition_id	Condition_name	Severity_level	Patient_id	Treatment_id	Doctor_id
▶	1	Heart Disease	Severe	101	201	1
	1	Heart Disease	Severe	101	201	5
	6	Coronary Artery Disease	Severe	106	206	1

10.Union :

Union of Doctors and Patients (similar column types).

```
select name as individual_name, 'patient' as role, doctor_id
from patients
union
select doctor_name as individual_name, 'doctor' as role, doctor_id
from doctors;
```

	Individual_Name	Role	Doctor_id
►	Raj Kumar	Patient	1
	Riya Patil	Patient	2
	Priyanka More	Patient	3
	Rajani Bapat	Patient	3
	Pooja Mane	Patient	5
	Kumar Sharma	Patient	1
	Alisha Patil	Patient	4
	Shyam Devan	Patient	5
	Karishma Sutar	Patient	4
	Dr.Anushree Patil	Doctor	1
	Dr.Sameer Shaikh	Doctor	2
	Dr.Sushant Davis	Doctor	3
	Dr.Aradhya Patel	Doctor	4
	Dr.Manas Devkar	Doctor	5

11.Joins :

To fetch all doctor id, Doctor name their specialization and patients id and name of patient which they are treated .

```
select d.doctor_id, d.doctor_name, d.specialization, p.patient_id, p.name as patient_name
from patients p right join doctors d on p.doctor_id = d.doctor_id;
```

	Doctor_id	Doctor_Name	Specialization	Patient_id	Patient_Name
►	1	Dr.Anushree Patil	Cardiology	101	Raj Kumar
	1	Dr.Anushree Patil	Cardiology	106	Kumar Sharma
	2	Dr.Sameer Shaikh	Neurology	102	Riya Patil
	3	Dr.Sushant Davis	Orthopedics	103	Priyanka More
	3	Dr.Sushant Davis	Orthopedics	104	Rajani Bapat
	4	Dr.Aradhya Patel	Oral surgery dentist	107	Alisha Patil
	4	Dr.Aradhya Patel	Oral surgery dentist	109	Karishma Sutar
	5	Dr.Manas Devkar	General Practitioner	105	Pooja Mane
	5	Dr.Manas Devkar	General Practitioner	108	Shyam Devan

Showing the number of patients per doctor, and the total cost of treatments for each doctor.

```
select d.doctor_name,  
count(p.patient_id) as number_of_patients,  
sum(t.cost) as total_treatment_cost  
from patients p  
inner join treatments t on p.patient_id = t.patient_id  
inner join doctors d on p.doctor_id = d.doctor_id  
group by d.doctor_name;
```

	Doctor_Name	Number_of_Patients	Total_Treatment_Cost
▶	Dr.Anushree Patil	2	172000.00
	Dr.Sameer Shaikh	1	25000.00
	Dr.Sushant Davis	2	83000.00
	Dr.Aradhya Patel	2	37000.00
	Dr.Manas Devkar	2	20000.00

12.Subqueries:

1.Single row Subqueries:

Find Treatment name and high cost that Cost More Than the Average Cost of All Treatments.

```
select Treatment_name,cost as High_Cost from treatments  
where cost>(select avg(cost) from treatments) order by cost desc;
```

	Treatment_name	High_Cost
▶	Heart Surgery	90000.00
	Angioplasty	82000.00
	Knee replacement	65000.00

Display Patient_name, Gender, Admission date and discharge date who treated by Dr. Anushree Patil.

```
select name as patient_name, gender, admission_date, discharge_date
from patients
where doctor_id = (select doctor_id from doctors where doctor_name = 'Dr. Anushree Patil');
```

	Patient_Name	Gender	Admission_date	Discharge_date
▶	Raj Kumar	Male	2024-01-08	2024-01-20
	Kumar Sharma	Male	2024-03-19	2024-03-30

2. Multiple row subquery:

List of Doctors name with their specialization who are treating more than 1 patients.

```
select doctor_name, specialization
from doctors
where doctor_id in (select doctor_id from patients group by doctor_id having count(patient_id) > 1);
```

	Doctor_Name	Specialization
▶	Dr. Anushree Patil	Cardiology
	Dr. Sushant Davis	Orthopedics
	Dr. Aradhya Patel	Oral surgery dentist
	Dr. Manas Devkar	General Practitioner

3. Multiple column subquery :

Display Highest-paid doctor for each specialization.

```
select doctor_name, specialization, salary
from doctors where (specialization, salary)
in ( select specialization, max(salary) from doctors group by specialization );
```

	doctor_name	specialization	salary
▶	Dr. Anushree Patil	Cardiology	120000.00
	Dr. Sameer Shaikh	Neurology	185000.00
	Dr. Sushant Davis	Orthopedics	100000.00
	Dr. Aradhya Patel	Oral surgery dentist	100000.00
	Dr. Manas Devkar	General Practitioner	100000.00

Find Medical Conditions for Patients Treated by Doctors with the Highest Salary and admitted before February 2024.

```
select condition_name
from medical_conditions
where patient_id in (select patient_id from patients where admission_date < '2024-02-28')
and doctor_id = (select doctor_id from doctors where salary = (select max(salary) from doctors));
```

	Condition_name
▶	Chronic Headache

13.VIEW:

create view : doctors information.

```
create view doctor_info as
select doctor_id, doctor_name, specialization, salary
from doctors;
select*from doctor_info;
```

	Doctor_id	Doctor_Name	Specialization	Salary
▶	1	Dr.Anushree Patil	Cardiology	120000.00
	2	Dr.Sameer Shaikh	Neurology	185000.00
	3	Dr.Sushant Davis	Orthopedics	100000.00
	4	Dr.Aradhya Patel	Oral surgery dentist	100000.00
	5	Dr.Manas Devkar	General Practitioner	100000.00

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